# MA087G Mathematical Statistics
# Beginner-Friendly Full Solution Guide
# 2026-01-07

## Source notes before solving

The official exam PDF was image-only. That means normal text extraction found no text, but the rendered page images were readable.

Two details are important:

- Uppgift 2 has an internal mismatch. The main displayed density is $f(x) = ax^2(2-x)$, but the printed fallback comment says $f(x) = 5x^3(2-x)/8$. I solve the main displayed density, because that is the actual formula in the question. I also give the fallback answers at the end of Uppgift 2.
- Uppgift 4b does not explicitly say that $X$ and $Y$ are independent. To calculate the variance of the taxi cost, we need independence. I use the standard exam assumption that they are independent.

## Uppgift 1

### What this question is really asking

We have two events, $A$ and $B$. An event just means something that may or may not happen.

The question gives three pieces of information:

$$P(A^c \cap B^c) = 0.60, \qquad P(A \cap B) = 0.20, \qquad P(A^c \cap B) = 0.10.$$

The little $c$ means "not". So:

$$A^c = \text{not } A, \qquad B^c = \text{not } B.$$

The question asks:

- Are $A$ and $B$ independent?
- What is $P(B \mid A^c)$, the probability that $B$ happens given that $A$ does not happen?

### How to recognize this type of question

The clues are:

- It talks about two events $A$ and $B$.
- It gives probabilities for combinations like $A^c \cap B$.
- It asks about independence.
- It asks for a conditional probability, shown by the vertical bar $|$.

So this is a two-event probability table problem. The safest beginner method is to fill in the four possible boxes:

$$A \cap B, \quad A \cap B^c, \quad A^c \cap B, \quad A^c \cap B^c.$$

### Tiny theory needed

The symbol $\cap$ means "and". For example:

$$A \cap B = A \text{ happens and } B \text{ happens.}$$

The symbol $P(B \mid A^c)$ means:

probability of $B$, if we already know that $A$ did not happen.

The formula for conditional probability is:

$$P(B \mid A^c) = \frac{P(B \cap A^c)}{P(A^c)}.$$

Independence means that knowing one event happened does not change the probability of the other. The exam formula for checking independence is:

$$P(A \cap B) = P(A)P(B).$$

### Formula and what everything means

For independence:

$$P(A \cap B) = P(A)P(B).$$

Here:

- $P(A \cap B)$ is the probability that both happen.
- $P(A)$ is the probability that $A$ happens.
- $P(B)$ is the probability that $B$ happens.
- If left side and right side are equal, they are independent.
- If they are not equal, they are not independent.

For conditional probability:

$$P(B \mid A^c) = \frac{P(B \cap A^c)}{P(A^c)}.$$

Here:

- The top is the part where $B$ happens and $A$ does not happen.
- The bottom is all cases where $A$ does not happen.

### Step-by-step solution

The four possible boxes must add to 1, because one of them must happen:

$$P(A \cap B) + P(A \cap B^c) + P(A^c \cap B) + P(A^c \cap B^c) = 1.$$

We already know three boxes:

$$P(A \cap B) = 0.20,$$
$$P(A^c \cap B) = 0.10,$$
$$P(A^c \cap B^c) = 0.60.$$

Find the missing box:

$$\begin{aligned} P(A \cap B^c) &= 1 - 0.20 - 0.10 - 0.60 \\ &= 1 - 0.90 \\ &= 0.10. \end{aligned}$$

This means the probability that $A$ happens but $B$ does not happen is 0.10.

Now find $P(A)$. Event $A$ happens in two boxes:

$$A \cap B$$

and

$$A \cap B^c.$$

So:

$$\begin{aligned} P(A) &= P(A \cap B) + P(A \cap B^c) \\ &= 0.20 + 0.10 \\ &= 0.30. \end{aligned}$$

Now find $P(B)$. Event $B$ happens in two boxes:

$$A \cap B$$

and

$$A^c \cap B.$$

So:

$$\begin{aligned} P(B) &= P(A \cap B) + P(A^c \cap B) \\ &= 0.20 + 0.10 \\ &= 0.30. \end{aligned}$$

Now check independence:

$$\begin{aligned} P(A)P(B) &= 0.30 \cdot 0.30 \\ &= 0.09. \end{aligned}$$

But:

$$P(A \cap B) = 0.20.$$

Since:

$$0.20 \neq 0.09,$$

$A$ and $B$ are not independent.

Now calculate $P(B \mid A^c)$:

$$P(B \mid A^c) = \frac{P(B \cap A^c)}{P(A^c)}.$$

The top is given:

$$P(B \cap A^c) = P(A^c \cap B) = 0.10.$$

The bottom $P(A^c)$ means all cases where $A$ does not happen:

$$\begin{aligned} P(A^c) &= P(A^c \cap B) + P(A^c \cap B^c) \\ &= 0.10 + 0.60 \\ &= 0.70. \end{aligned}$$

Therefore:

$$\begin{aligned} P(B \mid A^c) &= \frac{0.10}{0.70} \\ &= \frac{1}{7} \\ &\approx 0.143. \end{aligned}$$

### Answer

$$\boxed{A \text{ and } B \text{ are not independent}}$$

$$\boxed{P(B \mid A^c) = \frac{1}{7} \approx 0.143}$$

### What to memorize for the exam

For independence, always check:

$$P(A \cap B) = P(A)P(B).$$

For conditional probability, always use:

$$P(\text{wanted} \mid \text{given}) = \frac{P(\text{wanted and given})}{P(\text{given})}.$$

### Common trap

Do not compare $P(A \cap B)$ with $P(A) + P(B)$. Independence uses multiplication, not addition.

### Repeated exam pattern

This is the same broad pattern as 2025-08, Uppgift 1: two-event probability with complements and conditional probability. The numbers changed, but the solving method is the same.

# Uppgift 2

## What this question is really asking

We are given a function $f(x)$ and need to make it into a probability density function.

A density function is used for a continuous random variable. Continuous means $X$ can take values on an interval, like between 0 and 2, instead of only separate values like $0, 1, 2$.

The question asks us to:

- find the constant $a$,
- find the distribution function $F_X(x)$,
- calculate $P(X > 1)$,
- calculate expected value and standard deviation.

## How to recognize this type of question

The clues are:

- The question says "täthetsfunktion", which means density function.
- The function is given on an interval $0 \leq x \leq 2$.
- It asks for $F_X$, which is the distribution function/CDF.
- It asks for expected value and standard deviation.

So this is an integral problem. For continuous random variables, probabilities are areas, and areas are found with integrals.

## Tiny theory needed

The displayed density is:

$$f(x) = \begin{cases} ax^2(2-x), & 0 \leq x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

For $f$ to be a density, the total area under the curve must be 1:

$$\int_{-\infty}^{\infty} f(x)\,dx = 1.$$

Since $f(x) = 0$ outside $0 \leq x \leq 2$, we only integrate from 0 to 2.

The distribution function $F_X(x)$ means:

$$F_X(x) = P(X \leq x).$$

It tells us how much probability is to the left of $x$.

## Formula and what everything means

Density condition:

$$\int_0^2 f(x)\,dx = 1.$$

Here:

- $f(x)$ is the density.
- The integral is the area under $f(x)$.
- The total area must equal 1, because total probability is 100%.

CDF formula:

$$F_X(x) = \int_{-\infty}^{x} f(t)\,dt.$$

Here:

- $F_X(x)$ is probability up to $x$.
- $t$ is only a temporary integration letter.

Expected value:

$$E(X) = \int x f(x)\,dx.$$

This is the long-run average value.

Variance:

$$V(X) = E(X^2) - [E(X)]^2.$$

Standard deviation:

$$D(X) = \sqrt{V(X)}.$$

## Step-by-step solution

### Part a: Find $a$.

Use the density condition:

$$\int_0^2 ax^2(2-x)\,dx = 1.$$

First expand inside the integral:

$$x^2(2-x) = 2x^2 - x^3.$$

So:

$$a\int_0^2 (2x^2 - x^3)\,dx = 1.$$

Now integrate:

$$\int 2x^2\,dx = \frac{2x^3}{3}, \qquad \int x^3\,dx = \frac{x^4}{4}.$$

So:

$$a\left[\frac{2x^3}{3} - \frac{x^4}{4}\right]_0^2 = 1.$$

Insert the upper limit 2:

$$\begin{aligned}\frac{2\cdot 2^3}{3} - \frac{2^4}{4} &= \frac{16}{3} - 4 \\ &= \frac{16}{3} - \frac{12}{3} \\ &= \frac{4}{3}.\end{aligned}$$

Insert the lower limit 0, which gives 0. Therefore:

$$\begin{aligned}a \cdot \frac{4}{3} &= 1 \\ a &= \frac{3}{4}.\end{aligned}$$

**Part b: Find $F_X(x)$.**

The CDF must be piecewise:

- If $x < 0$, we are before the density starts, so $F_X(x) = 0$.
- If $0 \le x \le 2$, integrate from 0 to $x$.
- If $x > 2$, we have passed the whole density, so $F_X(x) = 1$.

For $0 \le x \le 2$:

$$\begin{aligned}F_X(x) &= \int_0^x \frac{3}{4}t^2(2-t)\,dt \\ &= \frac{3}{4}\int_0^x (2t^2 - t^3)\,dt \\ &= \frac{3}{4}\left[\frac{2t^3}{3} - \frac{t^4}{4}\right]_0^x \\ &= \frac{3}{4}\left(\frac{2x^3}{3} - \frac{x^4}{4}\right).\end{aligned}$$

Now multiply in $\frac{3}{4}$:

$$F_X(x) = \frac{x^3}{2} - \frac{3x^4}{16}.$$

Therefore:

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \dfrac{x^3}{2} - \dfrac{3x^4}{16}, & 0 \le x \le 2, \\ 1, & x > 2. \end{cases}$$

**Part c: Calculate $P(X > 1)$.**

The CDF gives $P(X \leq 1)$. We want the opposite side:

$$P(X > 1) = 1 - P(X \leq 1) = 1 - F_X(1).$$

Calculate $F_X(1)$:

$$\begin{aligned} F_X(1) &= \frac{1^3}{2} - \frac{3 \cdot 1^4}{16} \\ &= \frac{1}{2} - \frac{3}{16} \\ &= \frac{8}{16} - \frac{3}{16} \\ &= \frac{5}{16}. \end{aligned}$$

So:

$$\begin{aligned} P(X > 1) &= 1 - \frac{5}{16} \\ &= \frac{11}{16}. \end{aligned}$$

**Part d: Expected value and standard deviation.**

First calculate $E(X)$:

$$\begin{aligned} E(X) &= \int_0^2 x f(x)\, dx \\ &= \int_0^2 x \cdot \frac{3}{4} x^2 (2 - x)\, dx \\ &= \frac{3}{4} \int_0^2 x^3 (2 - x)\, dx \\ &= \frac{3}{4} \int_0^2 (2x^3 - x^4)\, dx. \end{aligned}$$

Now integrate:

$$\begin{aligned} E(X) &= \frac{3}{4} \left[ \frac{x^4}{2} - \frac{x^5}{5} \right]_0^2 \\ &= \frac{3}{4} \left( \frac{2^4}{2} - \frac{2^5}{5} \right) \\ &= \frac{3}{4} \left( 8 - \frac{32}{5} \right) \\ &= \frac{3}{4} \cdot \frac{8}{5} \\ &= \frac{6}{5}. \end{aligned}$$

Now calculate $E(X^2)$, because variance needs it:

$$\begin{aligned} E(X^2) &= \int_0^2 x^2 f(x)\,dx \\ &= \int_0^2 x^2 \cdot \frac{3}{4}x^2(2-x)\,dx \\ &= \frac{3}{4}\int_0^2 (2x^4 - x^5)\,dx \\ &= \frac{3}{4}\left[\frac{2x^5}{5} - \frac{x^6}{6}\right]_0^2 \\ &= \frac{3}{4}\left(\frac{64}{5} - \frac{64}{6}\right) \\ &= \frac{3}{4}\left(\frac{64}{5} - \frac{32}{3}\right) \\ &= \frac{8}{5}. \end{aligned}$$

Now variance:

$$\begin{aligned} V(X) &= E(X^2) - [E(X)]^2 \\ &= \frac{8}{5} - \left(\frac{6}{5}\right)^2 \\ &= \frac{8}{5} - \frac{36}{25} \\ &= \frac{40}{25} - \frac{36}{25} \\ &= \frac{4}{25}. \end{aligned}$$

Standard deviation:

$$\begin{aligned} D(X) &= \sqrt{V(X)} \\ &= \sqrt{\frac{4}{25}} \\ &= \frac{2}{5}. \end{aligned}$$

## Answer

For the displayed density $ax^2(2-x)$:

$$\boxed{a = \frac{3}{4}}$$

$$\boxed{F_X(x) = \begin{cases} 0, & x < 0, \\ \frac{x^3}{2} - \frac{3x^4}{16}, & 0 \le x \le 2, \\ 1, & x > 2 \end{cases}}$$

$$\boxed{P(X > 1) = \frac{11}{16}}$$

$$\boxed{E(X) = \frac{6}{5}}, \qquad \boxed{D(X) = \frac{2}{5}}.$$

If the printed fallback density $5x^3(2-x)/8$ was intended instead, then:

$$F_X(x) = \frac{5x^4}{16} - \frac{x^5}{8}, \qquad P(X > 1) = \frac{13}{16},$$

$$E(X) = \frac{4}{3}, \qquad D(X) = \sqrt{\frac{8}{63}}.$$

### What to memorize for the exam

For density questions:

- First make total area equal to 1.
- Then find the CDF by integrating the density.
- For $P(X > c)$, use $1 - F(c)$.
- For standard deviation, calculate variance first.

### Common trap

Do not write only the middle part of the CDF. A CDF must also say what happens before and after the interval.

### Repeated exam pattern

This is the same pattern as 2025-08, Uppgift 2: continuous density normalization, CDF, probability, expected value, and standard deviation. The density changed, but the solving recipe is identical.

## Uppgift 3

### What this question is really asking

We have two independent exponential random variables:

$$X \sim \text{Exp}(0.04), \qquad Y \sim \text{Exp}(0.05).$$

The new variable is:

$$Z = \min(X, Y).$$

That means $Z$ is whichever of $X$ and $Y$ happens first/smaller.

The question asks:

- What distribution does $Z$ have?
- What is $P(Z \geq 10)$?
- What is $E(Z)$?

### How to recognize this type of question

The clues are:

- It says $X \in \text{Exp}(0.04)$ and $Y \in \text{Exp}(0.05)$.
- It says independent, which means we may multiply probabilities.
- It defines $Z = \min(X, Y)$.

For a minimum, use survival probabilities. Survival means probability of being larger than a value.

### Tiny theory needed

If $X \sim \text{Exp}(\lambda)$, then:

$$P(X > t) = e^{-\lambda t}.$$

Here $\lambda$ is the rate.

For $Z = \min(X, Y)$, the event $Z > t$ means both variables are bigger than $t$:

$$Z > t$$

means:

$$X > t \text{ and } Y > t.$$

Because $X$ and $Y$ are independent, we can multiply:

$$P(X > t \text{ and } Y > t) = P(X > t)P(Y > t).$$

### Formula and what everything means

Exponential survival formula:

$$P(X > t) = e^{-\lambda t}.$$

Here:

- $t$ is the time or value we compare against.
- $\lambda$ is the exponential rate.
- $e^{-\lambda t}$ gives the probability that the variable is still above $t$.

For a minimum of independent exponentials:

$$\min(\text{Exp}(\lambda_1), \text{Exp}(\lambda_2)) \sim \text{Exp}(\lambda_1 + \lambda_2).$$

In words: for the minimum, the rates add.

### Step-by-step solution

**Part a: Find the distribution of $Z$.**

Start with:

$$Z = \min(X, Y).$$

For $z \geq 0$:

$$P(Z > z) = P(\min(X, Y) > z).$$

The minimum is greater than $z$ only if both $X$ and $Y$ are greater than $z$:

$$P(Z > z) = P(X > z,\ Y > z).$$

Since $X$ and $Y$ are independent:

$$P(Z > z) = P(X > z)P(Y > z).$$

Use the exponential survival formula:

$$P(X > z) = e^{-0.04z},$$
$$P(Y > z) = e^{-0.05z}.$$

Multiply:

$$\begin{aligned} P(Z > z) &= e^{-0.04z}e^{-0.05z} \\ &= e^{-0.09z}. \end{aligned}$$

This is the survival function of an exponential variable with rate 0.09. Therefore:

$$Z \sim \text{Exp}(0.09).$$

If we want the distribution function:

$$F_Z(z) = P(Z \leq z).$$

Since:

$$P(Z \leq z) = 1 - P(Z > z),$$

we get:

$$F_Z(z) = \begin{cases} 0, & z < 0, \\ 1 - e^{-0.09z}, & z \geq 0. \end{cases}$$

**Part b: Calculate $P(Z \geq 10)$.**

For a continuous variable, $P(Z \geq 10)$ and $P(Z > 10)$ are the same, because the probability of exactly 10 is 0.

Use the survival function:

$$\begin{aligned} P(Z \geq 10) &= P(Z > 10) \\ &= e^{-0.09 \cdot 10} \\ &= e^{-0.9} \\ &\approx 0.407. \end{aligned}$$

**Part c: Find $E(Z)$.**

For an exponential variable:

$$E(Z) = \frac{1}{\lambda}.$$

Here $\lambda = 0.09$, so:

$$\begin{aligned} E(Z) &= \frac{1}{0.09} \\ &= \frac{100}{9} \\ &\approx 11.11. \end{aligned}$$

### Answer

$$\boxed{Z \sim \text{Exp}(0.09)}$$

$$\boxed{P(Z \geq 10) = e^{-0.9} \approx 0.407}$$

$$\boxed{E(Z) = \frac{100}{9} \approx 11.11}$$

### What to memorize for the exam

For independent exponential variables:

$$\min(X, Y) \sim \text{Exp}(\lambda_X + \lambda_Y).$$

Minimum means the first one to happen, so the rates add.

### Common trap

Do not add the expected values of $X$ and $Y$. The minimum is not $X + Y$. For the minimum, add the rates, not the means.

### Repeated exam pattern

No earlier matching pattern found yet in the processed folders. Track this as: minimum of independent exponential variables.

## Uppgift 4

### What this question is really asking

This question has two separate parts.

Part a is about a discrete random variable. Discrete means it can only take a few separate values:

$$0, \quad 50, \quad 100.$$

We need its distribution, expected value, and standard deviation.

Part b is about a taxi cost built from two normally distributed variables:

$$X = \text{distance in km}, \qquad Y = \text{time in minutes}.$$

We must find the probability that the total cost is between 150 kr and 200 kr.

### How to recognize this type of question

Part a clues:

- The possible outcomes are fixed amounts of money.
- The text gives percentages and ratio information.
- It asks for expected value and standard deviation.

Part b clues:

- The question says $X \in N(3, 1)$ and $Y \in N(8, 2)$.
- That means normal distribution.
- The cost is a linear expression: start fee plus km fee plus minute fee.

### Tiny theory needed

For a discrete variable:

$$E(X) = \sum xP(X = x).$$

This means:

multiply each value by its probability, then add.

For variance:

$$V(X) = E(X^2) - [E(X)]^2.$$

Then:

$$D(X) = \sqrt{V(X)}.$$

For a linear combination of independent normal variables, the result is also normal.

If:

$$C = 57 + 14X + 9Y,$$

then:

$$E(C) = 57 + 14E(X) + 9E(Y).$$

For variance:

$$V(C) = 14^2V(X) + 9^2V(Y).$$

We square the coefficients in variance.

### Formula and what everything means

Discrete expected value:

$$E(X) = \sum xP(X = x).$$

Here:

- $x$ is a possible value.
- $P(X = x)$ is the probability of that value.
- The sum means add all value-times-probability terms.

Normal standardization:

$$z = \frac{x - \mu}{\sigma}.$$

Here:

- $x$ is the boundary value we are checking.
- $\mu$ is the mean/center.
- $\sigma$ is the standard deviation/spread.
- $z$ tells how many standard deviations away from the mean we are.

### Step-by-step solution

**Part a: Distribution, expected value, and standard deviation.**

The contribution can be:

$$0, \quad 50, \quad 100.$$

The question says 46% gave nothing:

$$P(X = 0) = 0.46.$$

So the remaining probability is:

$$1 - 0.46 = 0.54.$$

This 0.54 is split between 50 kr and 100 kr.

The share giving 50 kr is twice as large as the share giving 100 kr. Let:

$$P(X = 100) = q.$$

Then:

$$P(X = 50) = 2q.$$

Together they must add to 0.54:

$$\begin{aligned} q + 2q &= 0.54 \\ 3q &= 0.54 \\ q &= 0.18. \end{aligned}$$

So:

$$P(X = 100) = 0.18,$$

and:

$$P(X = 50) = 2q = 2 \cdot 0.18 = 0.36.$$

The distribution is:

| $x$ | 0 | 50 | 100 |
|---|---|---|---|
| $P(X = x)$ | 0.46 | 0.36 | 0.18 |

Now expected value:

$$\begin{aligned} E(X) &= 0 \cdot 0.46 + 50 \cdot 0.36 + 100 \cdot 0.18 \\ &= 0 + 18 + 18 \\ &= 36. \end{aligned}$$

So the average contribution is 36 kr.

Now calculate $E(X^2)$:

$$\begin{aligned} E(X^2) &= 0^2 \cdot 0.46 + 50^2 \cdot 0.36 + 100^2 \cdot 0.18 \\ &= 0 + 2500 \cdot 0.36 + 10000 \cdot 0.18 \\ &= 0 + 900 + 1800 \\ &= 2700. \end{aligned}$$

Variance:

$$\begin{aligned} V(X) &= E(X^2) - [E(X)]^2 \\ &= 2700 - 36^2 \\ &= 2700 - 1296 \\ &= 1404. \end{aligned}$$

Standard deviation:

$$\begin{aligned} D(X) &= \sqrt{1404} \\ &\approx 37.47. \end{aligned}$$

**Part b: Taxi cost probability.**

The cost has:

- 57 kr start fee,
- 14 kr per km,
- 9 kr per minute.

So the total cost is:

$$C = 57 + 14X + 9Y.$$

We use the standard convention $N(\mu, \sigma)$, so:

$$X \sim N(3, 1), \qquad Y \sim N(8, 2).$$

This means:

$$E(X) = 3, \quad D(X) = 1,$$

and:

$$E(Y) = 8, \quad D(Y) = 2.$$

Find the mean of $C$:

$$\begin{aligned} E(C) &= 57 + 14E(X) + 9E(Y) \\ &= 57 + 14 \cdot 3 + 9 \cdot 8 \\ &= 57 + 42 + 72 \\ &= 171. \end{aligned}$$

Now find the variance of $C$. We assume $X$ and $Y$ are independent:

$$V(C) = 14^2 V(X) + 9^2 V(Y).$$

Since standard deviation of $X$ is 1, variance is $1^2 = 1$. Since standard deviation of $Y$ is 2, variance is $2^2 = 4$:

$$\begin{aligned} V(C) &= 14^2 \cdot 1 + 9^2 \cdot 4 \\ &= 196 + 324 \\ &= 520. \end{aligned}$$

So:

$$D(C) = \sqrt{520} \approx 22.80.$$

Therefore:

$$C \sim N(171, \sqrt{520}).$$

We want:

$$P(150 \leq C \leq 200).$$

Standardize both endpoints:

$$\begin{aligned} z_1 &= \frac{150 - 171}{\sqrt{520}} \\ &\approx -0.921, \end{aligned}$$

and

$$\begin{aligned} z_2 &= \frac{200 - 171}{\sqrt{520}} \\ &\approx 1.272. \end{aligned}$$

So:

$$\begin{aligned} P(150 \leq C \leq 200) &= P(-0.921 \leq Z \leq 1.272) \\ &= \Phi(1.272) - \Phi(-0.921) \\ &\approx 0.898 - 0.179 \\ &\approx 0.720. \end{aligned}$$

### Answer

| $x$ | 0 | 50 | 100 |
|---|---|---|---|
| $P(X = x)$ | 0.46 | 0.36 | 0.18 |

$$\boxed{E(X) = 36 \text{ kr}}, \qquad \boxed{D(X) \approx 37.47 \text{ kr}}$$

$$\boxed{P(150 \leq C \leq 200) \approx 0.720}$$

### What to memorize for the exam

For discrete variables:

$$E(X) = \sum xP(X = x), \qquad V(X) = E(X^2) - [E(X)]^2.$$

For linear normal combinations:

means add directly, variances add with squared coefficients.

### Common trap

Do not add standard deviations directly. In part b, $14X + 9Y$ means the variance uses $14^2$ and $9^2$, not 14 and 9.

### Repeated exam pattern

Part a is a discrete-distribution-from-word-proportions pattern. Part b is the same linear-combination idea as 2025-08, Uppgift 3c, and the normal linear combination pattern tracked from this exam.

## Uppgift 5

### What this question is really asking

This is a maximum likelihood question, often shortened to ML.

The random variable $X$ counts how many successful free throws happen before the first missed free throw.

The probability function is:

$$p_X(k) = p^k(1 - p), \qquad k = 0, 1, 2, \ldots$$

The unknown number is $p$, the probability of making one free throw.

The question asks:

- Derive the ML estimate of $p$.
- Use the observed data to calculate the estimate.

### How to recognize this type of question

The clues are:

- It says "ML-skattning", meaning maximum likelihood estimate.
- A probability function with an unknown parameter $p$ is given.
- A sample $x_1, \ldots, x_n$ is given.

So we use the ML routine:

$$\text{likelihood} \to \text{log-likelihood} \to \text{differentiate} \to \text{solve}.$$

### Tiny theory needed

The likelihood is the probability of seeing the data, treated as a function of the unknown parameter $p$.

Because the observations are independent, we multiply their probabilities:

$$L(p) = \prod_{i=1}^{n} p^{x_i}(1-p).$$

We take logs because products are hard to differentiate, but logs turn products into sums.

### Formula and what everything means

Probability function:

$$p_X(k) = p^k(1-p).$$

Here:

- $k$ is the number of successes before the first miss.
- $p$ is the probability of success on one shot.
- $1-p$ is the probability of the first miss.

Likelihood:

$$L(p) = \prod_{i=1}^{n} p^{x_i}(1-p).$$

Here:

- $x_i$ is one observed result.
- $n$ is the number of observations.
- $\prod$ means multiply all the terms.

### Step-by-step solution

**Part a: Derive the ML estimate.**

Start with the likelihood:

$$L(p) = \prod_{i=1}^{n} p^{x_i}(1-p).$$

All the powers of $p$ combine:

$$p^{x_1}p^{x_2}\cdots p^{x_n} = p^{x_1+x_2+\cdots+x_n}.$$

Let:

$$S = \sum_{i=1}^{n} x_i.$$

This means $S$ is the sum of all observed values.

There is one $(1-p)$ factor for each observation, so there are $n$ of them:

$$(1-p)^n.$$

Therefore:

$$L(p) = p^S(1-p)^n.$$

Now take the natural logarithm:

$$\begin{aligned}\ell(p) &= \ln L(p) \\ &= \ln\left(p^S(1-p)^n\right).\end{aligned}$$

Use log rules:

$$\ell(p) = S\ln p + n\ln(1-p).$$

Now differentiate:

$$\frac{d}{dp}(S\ln p) = \frac{S}{p}.$$

Also:

$$\frac{d}{dp}\big(n\ln(1-p)\big) = -\frac{n}{1-p}.$$

So:

$$\ell'(p) = \frac{S}{p} - \frac{n}{1-p}.$$

Set the derivative equal to zero:

$$\frac{S}{p} - \frac{n}{1-p} = 0.$$

Move the second term to the other side:

$$\frac{S}{p} = \frac{n}{1-p}.$$

Cross-multiply:

$$S(1-p) = np.$$

Expand the left side:

$$S - Sp = np.$$

Move the $p$-terms to the right:

$$\begin{aligned} S &= np + Sp \\ S &= p(n + S). \end{aligned}$$

Solve for $p$:

$$\hat{p}_{ML} = \frac{S}{S+n}.$$

Since $S = \sum x_i$, we can write:

$$\hat{p}_{ML} = \frac{\sum_{i=1}^{n} x_i}{\sum_{i=1}^{n} x_i + n}.$$

**Part b: Insert the data.**

The observed data are:

$$2,\ 0,\ 9,\ 1,\ 6,\ 7,\ 5,\ 3.$$

There are:

$$n = 8$$

observations.

Add the observations:

$$\begin{aligned} S &= 2 + 0 + 9 + 1 + 6 + 7 + 5 + 3 \\ &= 33. \end{aligned}$$

Now insert $S = 33$ and $n = 8$:

$$\begin{aligned} \hat{p}_{ML} &= \frac{33}{33+8} \\ &= \frac{33}{41} \\ &\approx 0.805. \end{aligned}$$

**Answer**

$$\boxed{\hat{p}_{ML} = \frac{\sum_{i=1}^{n} x_i}{\sum_{i=1}^{n} x_i + n}}$$

$$\boxed{\hat{p}_{ML} = \frac{33}{41} \approx 0.805}$$

**What to memorize for the exam**

For ML questions, the routine is:

$$L \to \ln L \to \text{differentiate} \to \text{set equal to zero} \to \text{solve}.$$

For this specific distribution:

$$\hat{p} = \frac{\sum x_i}{\sum x_i + n}.$$

### Common trap

Do not stop after writing the likelihood. The estimate comes from differentiating the log-likelihood and solving.

### Repeated exam pattern

This is the same broad ML pattern as 2025-08, Uppgift 4. The distribution is different, but the workflow is the same.

## Uppgift 6

### What this question is really asking

This question is about a sample of daily parking income.

We are given:

$$n = 30, \qquad \bar{x} = 1260, \qquad s = 150.$$

Here:

- $n = 30$ means 30 weekdays were observed.
- $\bar{x} = 1260$ means the sample average was 1260 kr.
- $s = 150$ means the sample standard deviation was 150 kr.

The question asks for:

- a 90% confidence interval for the true mean daily income,
- hypotheses for testing the consultant's claim $\mu = 1280$,
- a hypothesis test at significance level 10%.

### How to recognize this type of question

The clues are:

- It says "konfidensintervall", so part a is a confidence interval.
- It gives a sample mean and sample standard deviation.
- It says the data are from a normal distribution.
- It asks whether a claim about the mean can be rejected, so parts b and c are a hypothesis test.

Because the population standard deviation $\sigma$ is not given, we use the $t$-distribution.

### Tiny theory needed

A confidence interval gives a range of plausible values for the true mean $\mu$.

Here we do not know the true standard deviation $\sigma$. We only know the sample standard deviation $s = 150$. That is why we use a $t$-value instead of a normal $z$-value.

A hypothesis test compares:

- $H_0$: the default claim,
- $H_1$: the alternative.

The consultant claimed:

$$\mu = 1280.$$

The question asks whether this claim is reasonable or can be rejected. Since it does not say specifically "larger" or "smaller", we use a two-sided test:

$$H_1 : \mu \neq 1280.$$

### Formula and what everything means

Confidence interval for mean with unknown $\sigma$:

$$\bar{x} \pm t_{\alpha/2,n-1} \frac{s}{\sqrt{n}}.$$

Here:

- $\bar{x}$ is the sample mean.
- $t_{\alpha/2,n-1}$ is the $t$-critical value.
- $n - 1$ is degrees of freedom.
- $s$ is the sample standard deviation.
- $s/\sqrt{n}$ is the standard error, the spread of the sample mean.

Test statistic:

$$T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

Here:

- $\mu_0$ is the value claimed in $H_0$, here 1280.
- $T$ tells how many standard errors the observed mean is from the claimed mean.

### Step-by-step solution

#### Part a: 90% confidence interval.

Given:

$$n = 30, \qquad \bar{x} = 1260, \qquad s = 150.$$

A 90% interval means:

$$\alpha = 0.10.$$

Because the interval has two sides, split this into two tails:

$$\alpha/2 = 0.05.$$

Degrees of freedom:

$$n - 1 = 30 - 1 = 29.$$

From a $t$-table:

$$t_{0.95,29} \approx 1.699.$$

Now calculate the standard error:

$$\begin{aligned} \frac{s}{\sqrt{n}} &= \frac{150}{\sqrt{30}} \\ &\approx \frac{150}{5.477} \\ &\approx 27.386. \end{aligned}$$

Now calculate the margin of error:

$$1.699 \cdot 27.386 \approx 46.53.$$

So the confidence interval is:

$$1260 \pm 46.53.$$

Lower endpoint:

$$1260 - 46.53 = 1213.47.$$

Upper endpoint:

$$1260 + 46.53 = 1306.53.$$

So:

$$(1213.47,\ 1306.53).$$

**Part b: State hypotheses.**

The consultant claimed:

$$\mu = 1280.$$

So the null hypothesis is:

$$H_0 : \mu = 1280.$$

The question asks whether the claim is reasonable or can be rejected. It does not say we only care about higher or lower, so use a two-sided alternative:

$$H_1 : \mu \neq 1280.$$

**Part c: Perform the hypothesis test.**

Use:

$$T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

Insert values:

$$\begin{aligned} T &= \frac{1260 - 1280}{150/\sqrt{30}} \\ &= \frac{-20}{27.386} \\ &\approx -0.730. \end{aligned}$$

At significance level 10%, with a two-sided test and 29 degrees of freedom, the critical values are:

$$\pm t_{0.95,29} \approx \pm 1.699.$$

The rejection regions are:

$$T < -1.699$$

or

$$T > 1.699.$$

Our value is:

$$T = -0.730.$$

This is between $-1.699$ and $1.699$:

$$-1.699 < -0.730 < 1.699.$$

So we do not reject $H_0$.

Plain-language check: the claimed value 1280 is inside the 90% confidence interval:

$$1213.47 < 1280 < 1306.53.$$

So the consultant's claim is plausible from this sample.

### Answer

$$\boxed{\text{90\% confidence interval for } \mu: \ (1213.47,\ 1306.53) \text{ kr}}$$

$$\boxed{H_0 : \mu = 1280, \qquad H_1 : \mu \neq 1280}$$

$$\boxed{T \approx -0.730}$$

$$\boxed{\text{Do not reject } H_0. \text{ The consultant's claim cannot be rejected.}}$$

### What to memorize for the exam

If the question gives $s$, the sample standard deviation, and does not give the true $\sigma$, use a $t$-method:

$$\bar{x} \pm t\frac{s}{\sqrt{n}}.$$

For a hypothesis test:

$$T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

### Common trap

Do not use a $z$-test just because the sample size is 30. The safer course-exam rule is: if $\sigma$ is unknown and you use $s$, use $t$.

### Repeated exam pattern

This is related to 2025-08, Uppgift 6 because both are hypothesis tests for a mean. The difference is that 2026-01 uses unknown $\sigma$ and a two-sided $t$-test, while 2025-08 uses known $\sigma$ and a one-sided $z$-test.

## Mini-cheatsheet from this exam

### Probability

Complement:

$$P(A^c) = 1 - P(A).$$

Conditional probability:

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}.$$

Independence:

$$P(A \cap B) = P(A)P(B).$$

### Continuous density

To make a density:

$$\int f(x)\,dx = 1.$$

CDF:

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt.$$

Right-tail probability:

$$P(X > c) = 1 - F_X(c).$$

Variance:

$$V(X) = E(X^2) - [E(X)]^2.$$

### Exponential distribution

If $X \sim \text{Exp}(\lambda)$:

$$P(X > t) = e^{-\lambda t}, \qquad E(X) = \frac{1}{\lambda}.$$

For independent exponentials:

$$\min(X, Y) \sim \text{Exp}(\lambda_X + \lambda_Y).$$

### Discrete variables

Expected value:

$$E(X) = \sum xP(X = x).$$

Second moment:

$$E(X^2) = \sum x^2 P(X = x).$$

Variance:

$$V(X) = E(X^2) - [E(X)]^2.$$

### Linear combinations

Mean:

$$E(a + bX + cY) = a + bE(X) + cE(Y).$$

Variance, if independent:

$$V(a + bX + cY) = b^2V(X) + c^2V(Y).$$

### Normal standardization

$$z = \frac{x - \mu}{\sigma}.$$

### Maximum likelihood

ML workflow:

$$L \to \ln L \to \text{differentiate} \to \text{set equal to zero} \to \text{solve}.$$

For $p_X(k) = p^k(1 - p)$:

$$\hat{p} = \frac{\sum x_i}{\sum x_i + n}.$$

### Confidence interval and test for mean, unknown sigma

Confidence interval:

$$\bar{x} \pm t_{\alpha/2,n-1}\frac{s}{\sqrt{n}}.$$

Test statistic:

$$T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$